NEXASTORE

AI Machines • Developer Workstations • Next-Generation Computing



AI READY

Choosing a Laptop for Software Development

A practical guide for modern developers

NexaStore Knowledge Series • 2026 Edition

1. Development Workloads Are Memory-Hungry

Modern developers rarely run only an IDE. A realistic workstation may run an IDE, browser tabs, Docker containers, databases, local queues, cloud CLIs, test runners and AI coding tools at the same time. Memory capacity and sustained CPU performance therefore matter more than a thin chassis alone.

Development profile	RAM guidance	Storage guidance	GPU need
.NET / Java web developer	32 GB recommended	1 TB NVMe	Optional
Full-stack + containers	32-64 GB	1-2 TB NVMe	Optional
Microservices / local Kubernetes	64 GB preferred	2 TB NVMe	Optional
Data / ML developer	32-64+ GB	1-2 TB NVMe	Workload dependent
Local GenAI developer	64 GB preferred	2 TB+ NVMe	Discrete GPU strongly useful

2. What Matters for .NET Development

Area	Why it matters
CPU	Builds, tests, IDE analysis and container workloads benefit from modern multi-core performance.
RAM	Visual Studio, browsers, Docker, databases and local clusters can easily consume tens of gigabytes.
SSD	Fast random I/O improves package restore, builds, container images, databases and source operations.
Virtualization	Required for common container and local virtualization workflows.
Ports	External monitors, docks, Ethernet and test devices often matter in enterprise development.
Battery	Useful for mobility, but sustained builds and GPU workloads are best performed on AC power.

3. Example Developer Workstation

Layer	Example
IDE	Visual Studio / VS Code / JetBrains IDE
Runtime	.NET SDK, Node.js, Python where needed
Containers	Docker Desktop or compatible runtime
Local infrastructure	PostgreSQL, Redis, message broker, object storage emulator
Kubernetes	kubectl, Helm and local cluster when required
Cloud	Azure/AWS/GCP CLI and developer tooling
AI	Cloud model SDKs; optional local inference runtime

4. Developer + AI Scenario

A developer building RAG or agentic applications may simultaneously run the web application, API, PostgreSQL, vector extensions, Redis, an ingestion worker, containers and a local model runtime. In that scenario, 64 GB system memory and a discrete GPU can materially improve the experience.

Scenario	Recommended direction
Cloud-only AI APIs	Strong developer laptop; discrete GPU not required
Occasional small local models	32-64 GB RAM; integrated or entry discrete GPU may suffice
Regular local LLM work	64 GB RAM and higher-VRAM discrete GPU
CUDA-based ML / CV	NVIDIA-class discrete GPU and compatible software stack
Heavy local training	Workstation or cloud GPU rather than thin laptop

5. Buying Checklist

• 32 GB RAM minimum for a comfortable professional baseline; consider 64 GB for containers/AI.

• 1 TB NVMe minimum; 2 TB is preferable when storing container images and local models.

• Verify memory upgradeability before assuming it can be expanded later.

• Check virtualization support and operating-system compatibility.

• For local AI, compare GPU VRAM and software compatibility rather than relying on 'AI PC' branding alone.

• For enterprise use, consider warranty, docking, webcam, security and serviceability.

Educational notice

This guide is part of the fictional NexaStore educational knowledge base.